LAB 3

Acid-Base Titration: Molarity of an Unknown HCl

Lab handout and data sheet. Read it before lab. Goggles.

Lab	Tue 10/20/2026, 6:00 to 8:50 pm, Science Hall S-118 (Tuesday 6:00 pm section)
Report due	Tue 10/27/2026, start of lab (6:00 pm). Attach a copy of your data sheet.
Instructor	Paul Hendricks · phendricks@sciotocc.edu · Science Department office (614) 555-0166
Bring	Goggles, lab manual, calculator and this handout. The data sheet is the last page.

Purpose

Determine the molarity of an unknown hydrochloric acid solution by titration with standardized sodium hydroxide.

Background

$$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$$

The reaction is 1:1. At the equivalence point, mol NaOH added = mol HCl in the flask. The NaOH molarity is known and the pipet fixes the HCl volume, so the burette volume at the endpoint gives the HCl molarity.

Phenolphthalein is colorless in acid and turns pink just past the equivalence point. The endpoint is the first faint pink that lasts 30 seconds. Faint pink, not magenta. For a strong acid titrated with a strong base, the pH rises slowly at first, jumps sharply near the equivalence point (pH 7 at 25 °C), then levels off in the basic range.

Materials (per bench)

- 0.1000 M NaOH (standardized)
- Unknown HCl solution
- 25.00 mL volumetric pipet and pipet bulb
- 50 mL burette, ring stand, burette clamp, small funnel
- Three 250 mL Erlenmeyer flasks
- Phenolphthalein indicator, wash bottle of deionized water
- Waste beaker, white paper

Safety and waste

- Goggles on at all times in S-118, cleanup included.
- NaOH can damage eyes, even at 0.1 M. Skin contact: rinse with plenty of water and tell the instructor.
- Pipet bulb only. No mouth pipetting.
- Leftover NaOH goes in the labeled waste container on the front bench, never back in the stock bottle.

Procedure

1. Rinse the 50 mL burette with two 5 mL portions of 0.1000 M NaOH, draining each rinse through the tip into the waste beaker. Fill it, take the funnel out, run some NaOH through the tip and remove any air bubble.
2. Pipet a 25.00 mL aliquot of unknown HCl into a clean 250 mL Erlenmeyer flask. Add about 25 mL deionized water and 3 drops phenolphthalein. Set the flask on white paper.
3. Record the initial burette reading to 0.01 mL. Read the bottom of the meniscus at eye level. The initial reading does not have to be 0.00 mL. It has to be recorded.
4. Titrate while swirling. Near the endpoint the pink takes longer to fade: add NaOH drop by drop, then half drops (let a partial drop form, touch it to the flask wall, rinse it down with deionized water). Stop at the first faint pink that lasts 30 seconds.
5. Record the final burette reading to 0.01 mL.
6. Refill the burette and repeat steps 2 to 5 for three trials total. Each trial takes roughly 20 to 25 mL. Volumes delivered should agree within 0.10 mL. If they do not, run a fourth trial. Overshot to dark pink: note it on the data sheet and run another trial.
7. Cleanup: neutralized solutions go down the sink with running water. Rinse the burette and pipet with deionized water before you return them.

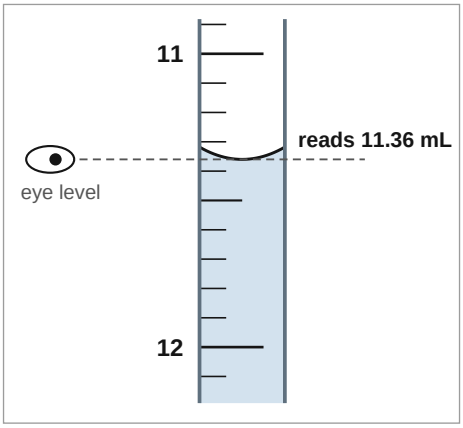


Figure 3.1. Meniscus between the 11.3 and 11.4 mL lines, a little past halfway. Record 11.36 mL.

Reading the burette

- Eye level with the bottom of the meniscus. No parallax.
- Lines every 0.1 mL. Estimate the second decimal: 11.36 mL.
- Numbers increase downward: the final reading is the larger one.
- Funnel out, no air bubble in the tip, no hanging drop.
- Write each reading on the data sheet as you take it.

From Lab 2 grading

- Volumes reported to 0.1 mL. A burette reads to 0.01 mL.
- Report table did not match the data sheet. Check transcription, data sheet → report.
- "Human error" is not an error source. Name the actual error.

Data and calculations

Record readings on the data sheet (last page) as you take them. Table 3.1 in your report uses the same columns, every reading and volume to 0.01 mL. Hand in a copy of the data sheet with the report.

V(NaOH) = final reading - initial reading	*mL, each trial*
mol NaOH = M(NaOH) × V(NaOH)	*V in liters*
mol HCl = mol NaOH	*1:1 mole ratio*
M(HCl) = mol HCl / 0.02500 L	*25.00 mL aliquot*

Report M(HCl) for each trial and the mean, 4 significant figures. Do not round between steps.

Titration curve

Sketch pH (y-axis) vs. mL of NaOH added (x-axis). No pH meter in this lab: draw the expected shape for a strong acid titrated with a strong base and label the equivalence point at your mean endpoint volume.

Report (20 points)

Section	What is graded	Points
Format and prelab	Title, purpose, brief procedure in past tense. Tables numbered, units in column headers. Prelab done before lab.	5
Data and tables	Table 3.1: initial reading, final reading and volume delivered for every trial, to 0.01 mL, matching the data sheet. Copy of the data sheet attached.	5
Calculations	One sample calculation with units. M(HCl) for each trial and the mean, 4 significant figures.	5
Results and discussion	Mean M(HCl). Titration curve. Two specific error sources and which way each one moves M(HCl). "Human error" is not an error source.	5
Total		**20**

Prelab

Answer before lab on your own paper. Checked at the start of lab.

1. Why is the burette rinsed with the NaOH solution, not only with deionized water, right before it is filled?
2. About 25 mL of deionized water goes into each flask. Why does that not change the volume of NaOH needed?
3. A trial ends with a dark pink flask. What happened, and what do you do?

Bench partners may share raw data. Calculations, tables and discussion are individual work.

Lab 3 Data Sheet

CHEM 1211 · Tue 10/20/2026 · Science Hall S-118

Acid-Base Titration: Molarity of an Unknown HCl

Name _____ **Bench partner** _____ **Date** _____

NaOH (titrant)	0.1000 M, standardized
HCl aliquot	25.00 mL, volumetric pipet
Indicator	Phenolphthalein, 3 drops. Endpoint: first faint pink that lasts 30 seconds.

Table 3.1. Titration data. Every reading and volume to 0.01 mL.

Trial	Initial reading (mL)	Final reading (mL)	Volume NaOH delivered (mL)
1			
2			
3			
4 (if needed)			

Volumes agree within 0.10 mL? Circle one: **yes** / **no**. If no, run trial 4.

Endpoint notes (overshoot, air bubble, spill, anything you changed):

Write readings here first, as you take them. Copy them into Table 3.1 of your report and check each one against this sheet. Hand in a copy of this sheet with the report.